

Exploring Information Technology Career Paths: Information Security Analyst vs. Network and  
Computer Systems Administrator

Phi Anh Kiet Nguyen

Department of Information Technology, Northern Virginia Community College

ITE 152: Introduction to CyberSecurity and Information Technology

Dr. Dave Filer

September 23, 2026

## **Exploring Information Technology Career Paths: Information Security Analyst vs. Network and Computer Systems Administrator**

The rapid evolution of digital infrastructure has transformed information technology (IT) into one of the most vital sectors of the global economy. As organizations increasingly depend on cloud computing, data analytics, and interconnected networks, the demand for skilled IT professionals continues to surge. To gain a deeper understanding of potential career opportunities in this field, I conducted extensive research using web-based resources, primarily the Department of Labor's Occupational Outlook Handbook (OOH) provided through the Northern Virginia Community College (NVCC) Library network. During my research, I evaluated various specializations—including software development, database administration, and system architecture—before narrowing my focus to two specific roles: Information Security Analyst and Network and Computer Systems Administrator. My initial interest in these career paths was sparked by a fascination with how modern enterprises balance seamlessly open accessibility for employees with robust defense mechanisms against persistent cyber threats.

### **Career Path 1: Information Security Analyst**

An Information Security Analyst plans, implements, and monitors security measures to protect an organization's computer networks and systems. Core responsibilities include conducting penetration testing<sup>1</sup>, analyzing network traffic for anomalies using Security Information and Event Management (SIEM) tools, responding to cybersecurity incidents, and establishing disaster recovery plans. Essential technical skills range from firewall configuration and vulnerability assessment to scriptwriting, alongside soft skills like analytical critical thinking and clear communication.

---

<sup>1</sup> Penetration testing refers to authorized simulated cyberattacks performed on a computer system to evaluate its security posture.

Entering this field typically requires a Bachelor's degree in Information Technology, Computer Science, Cybersecurity, or a related field. While state licensing is generally not mandated, industry-recognized examinations and professional certifications are critical for employment and career progression. Entry-level practitioners often obtain the CompTIA Security+ certification, while experienced analysts pursue advanced credentials such as Certified Information Systems Auditor (CISA) or Certified Information Systems Security Professional (CISSP).

According to the U.S. Bureau of Labor Statistics (2026), the median annual wage for information security analysts is approximately \$120,360 per year (or \$57.87 per hour). Higher levels of formal education, specialized certifications, and practical experience significantly influence compensation, with top earners in federal government and finance sectors exceeding \$174,000 annually.

The primary benefits of this career path include exceptional job stability, above-average earning potential, and the high-demand nature of the field, which offers extensive opportunities for remote work. Conversely, notable drawbacks include high workplace stress, the requirement to be on-call for emergency incident response, and potential burnout due to the relentless nature of cyber threats.

Opportunities for advancement in cybersecurity are abundant. Analysts can progress into senior roles such as Lead Security Architect, Cybersecurity Manager, or Chief Information Security Officer (CISO). Advancement typically requires higher-level technical certifications, expanded supervisory duties, and continuous education to remain ahead of sophisticated cyber attack vectors.

A typical day in the life of an Information Security Analyst begins with reviewing security logs and automated threat alerts generated overnight. The analyst investigates suspicious login

attempts or malware detection flags, coordinates with the network engineering team to patch software vulnerabilities, attends a risk assessment meeting, and ends the day updating the organization's incident response documentation.

Aspects of this career path that I find highly appealing include the dynamic, problem-solving environment and the vital role analysts play in protecting critical infrastructure. The primary aspect I find less appealing is the high level of stress associated with managing active security breaches under strict time constraints.

### **Career Path 2: Network and Computer Systems Administrator**

Network and Computer Systems Administrators are responsible for the day-to-day operation, installation, and maintenance of an organization's local area networks (LAN), wide area networks (WAN), intranets, and data communication systems. Key tasks include installing network hardware (routers and switches), configuring system software, monitoring network performance, performing data backups, and troubleshooting connectivity issues for end-users. Strong technical knowledge in operating systems (Linux, Windows Server) and hardware diagnostics is essential.

Educational requirements generally mandate a Bachelor's degree in Information Technology or Network Administration, although some employers accept an Associate degree combined with relevant work experience and technical certifications. Although state licensure is not required, professional certifications such as Cisco Certified Network Associate (CCNA), Microsoft Certified: Azure Administrator, or CompTIA Network+ are widely expected by employers.

Data from the U.S. Bureau of Labor Statistics (2026) indicates that the median annual wage for network and computer systems administrators is approximately \$95,360 per year (or \$45.85 per

hour). Advanced degrees or specialized vendor certifications directly correlate with higher earning brackets, particularly in telecommunications and enterprise technology management. Benefits of this role include a solid foundational career path in core infrastructure, predictable project cycles, and versatile skills applicable across almost every industry. Drawbacks include occasional scheduled weekend work for system maintenance, repetitive troubleshooting tasks, and physical demands related to rack-mounting server hardware.

Advancement opportunities include moving into positions such as Computer Network Architect, IT Operations Director, or Systems Engineer. Advancement generally requires ongoing training in cloud environments (AWS, Azure), managing larger infrastructure teams, and acquiring advanced architectural certifications. Continuing education is imperative as organizations shift toward hybrid cloud <sup>2</sup>environments.

A typical day for a Network Administrator involves monitoring network bandwidth usage, upgrading server firmware during scheduled downtime windows, resolving technical support tickets escalated from the help desk, and configuring access permissions for newly onboarded staff.

The most appealing aspect of this position is the satisfaction of building and maintaining tangible, stable hardware infrastructure that enables an entire organization to function efficiently. The less appealing aspect is the routine nature of user-level troubleshooting and the physical upkeep of hardware racks.

## **Conclusion**

Comparing both career paths reveals that while both require strong analytical skills and continuous technical education, their core focus areas differ significantly. Network

---

<sup>2</sup> Hybrid cloud environments combine on-premise infrastructure and private clouds with public cloud services such as AWS or Microsoft Azure.

administrators focus on constructing, optimizing, and maintaining operational availability, whereas Information Security Analysts focus on defending that infrastructure against deliberate external and internal threats. Information Security Analysts command higher average salaries (\$120,360 vs. \$95,360) and experience faster projected job growth due to escalating worldwide cyber attacks (U.S. Bureau of Labor Statistics, 2026).

Based on this research, I am most interested in pursuing a career as an Information Security Analyst. The combination of rapid industry growth, intellectual challenge, and the proactive nature of threat hunting aligns perfectly with my long-term professional aspirations in cybersecurity.

### Annotated Bibliography

Cybersecurity and Infrastructure Security Agency. (2026). *Cybersecurity workforce development and training resources*. U.S. Department of Homeland Security.

<https://www.cisa.gov/cybersecurity-workforce>

This government website provides comprehensive guidelines on cybersecurity career pathways, skill frameworks, and industry certifications. The resource is highly credible, current, and reliable, originating directly from the federal agency responsible for national cyber defense. It proved exceptionally useful for understanding official training expectations and mapping advancement opportunities within the Information Security Analyst path.

U.S. Bureau of Labor Statistics. (2026, April 17). *Information security analysts*. Occupational Outlook Handbook. U.S. Department of Labor.

<https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm>

This publication offers accurate, objective statistical data regarding job descriptions, median annual pay, educational requirements, and projected growth rates for information security analysts through 2032. As an official publication of the U.S. Department of Labor, the information is unbiased and authoritative. This resource served as the primary source for salary figures and educational prerequisites cited in the paper.

U.S. Bureau of Labor Statistics. (2026, April 17). *Network and computer systems administrators*. Occupational Outlook Handbook. U.S. Department of Labor.

<https://www.bls.gov/ooh/computer-and-information-technology/network-and-computer-systems-administrators.htm>

This official handbook entry outlines the duties, work environments, compensation metrics, and job prospects for network administrators. The source is highly valid, accurate, and

regularly updated by government economists. It provided essential comparative data needed to evaluate the daily responsibilities and advancement options for the second career path evaluated in this research paper.